

Soft Suave
The smart way..

Web Scraping

Solution For Logistics Business

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CASE STUDY BRIEF

- Our Client, one of the world's finest logistics management solution company, came to us for help.

- They had an existing web app platform and wanted us to implement a web scraping solution into it. This solution would extract data from a variety of dynamic sources, and scale up their business operations. Our developers with their high-end tech stack, efficiently tailored a web scraping solution and delivered a cost-effective service.

- Now, explore this case study to understand how we helped our client in their business operations through our solutions.

CLIENT OVERVIEW

- Our client is a cloud-based logistics management solution targeting businesses that ship goods, including manufacturers, retailers, and e-commerce companies. They were founded in 2018, and focus on cloud-based solutions with features like shipment tracking, automated alerts, contract rate management, and collaboration tools addressing the growing need for streamlined visibility and cost savings in freight transportation.

- Their target market includes businesses requiring simplified logistics processes. Their USP is established via their technology-driven approach and easy data integration across diverse systems.



REQUIREMENTS

- The client faced significant challenges in organizing their collected data. With data coming from various companies in different formats, checking its status was difficult. The inconsistent formats made storing and retrieving the data challenging.

- Initially, their web application used **ReactJS**, **NodeJS**, and **MongoDB**. However, MongoDB's limitations in scalability made it unable to handle large loads. When millions of data entries were loaded, the database struggled, leading to reduced web app performance and numerous user complaints.

- To address these issues, the client required a solution to improve the web app's performance and scalability. They also requested a web scraping solution to automatically run in the background, mining data from dynamic sources and consolidating it on a single platform.

Issue	Reason	Potential Losses
Unable to store **procured data** in an organized manner due to diverse format.	Limited scalability of existing **MongoDB** setup.	Reduced application performance leading to **user dissatisfaction** and **potential loss** of customers.

CHALLENGES FACED BY SOFT SUAVE

- **Overcoming Website Security:** Encountered several challenges while making the solution, such as dealing with CAPTCHA verification requirements on tracking sites. Our solution had to find ways to navigate these to keep the data flowing.

- **Adapting to Data Differences:** Managing variations in the structure of data across multiple tracking sites was another challenge. Our solution had to be flexible enough to handle these variations and collect the shipment data accurately.

SOLUTION OFFERED BY SOFT SUAVE

- Understanding the difficulties of data scraping in logistics, we designed an architectural solution to make it more manageable for logistics businesses.

- The web scraping solution had to be efficient, scalable, and fast. We thus implemented the solution using DynamoDB for scalability, and OpenSearch for fast & efficient data retrieval. With Puppeteer, the solution could mine data even without an existing user interface.



Tech Stack Used

Our developers decided on a highly powerful tech stack to make the solution easy to use, efficient, and cost-effective.

The tech stack utilized were,

Front-End

ReactJS

Back-End

NodeJS

Data Collection

Puppeteer

Data Search & Indexing

Open Search

Data Storage

MongoDB

DynamoDB



Key Features Of The Solution

Cost-Effective

We utilized AWS Lambda, a server that operates on a pay-as-you-go model, ensuring you only pay for what you consume.



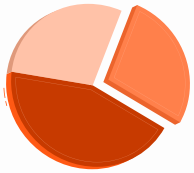
Lightning-Fast Data Retrieval

By utilizing DynamoDB, we ensured swift and efficient access to your data.



Improved Data Organization

The new solution standardizes data formats from all sources, streamlining storage, retrieval & analysis.



Benefits Of The Solution

- Our solution removes the necessity to manually inspect each carrier's website, which saves the users a lot of time and resources.
- Users obtain a unified and current perspective of all active shipments from various transporters on one platform.
- Users can get the latest information about their shipment status right away, which allows them to make decisions ahead of time.

By using the web scraping tool we created, the client's users can get better oversight and management of their multi-carrier freight network, which can make their operations more efficient, improve service quality, and maybe reduce expenses.



Maintenance & Support

We provided ongoing maintenance and support to ensure that the web app and web scraping tool performed well. We adapted our system to accommodate changes in carriers' (tracking site) UIs as they upgraded or modified their platforms.

RESULT

The web scraping solution we implemented greatly boosted performance and scalability, effectively tackling the client's challenges. We achieved efficient data retrieval and processing through the powerful new tech stack, providing users with timely and accurate insights for better logistics management. Our cutting-edge solution simplified operations, empowering clients to achieve their business objectives efficiently.



About Soft Suave

Soft Suave is a prominent IT outsourcing company with a rich history of serving global clients for more than eleven years. With our core expertise in IT staff augmentation, we assist businesses in augmenting their existing workforce by providing skilled professionals. Our team consists of talented, knowledgeable, and dynamic developers who are devoted to offering revolutionary and effective technology solutions. We offer tailored solutions that meet the unique needs of our clients, and our customer-centric approach ensures that our clients are involved at every stage of the project.

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